

Continuation-In-Part (CIP) Strategy: Holographic Micro-LED Display

Project: HEALTH-BAND Neuro **Base Application:** US Provisional App. No. 64/073,334
(Filed May 23, 2026) **CIP Filing Deadline:** May 23, 2027

1. Objective

The objective of this Continuation-In-Part (CIP) application is to expand the claims of the original HEALTH-BAND Neuro provisional patent to include a novel display architecture. While the V1 hardware relies on a standard Micro OLED panel, the CIP will protect the future integration of a bezel-less, flexible holographic micro-LED display.

2. The Core CIP Claim

The primary claim to be added via the CIP application is as follows:

"A wearable health monitoring apparatus comprising a micro-LED array positioned behind a holographic diffuser film embedded flush in the outer surface of a flexible wristband, wherein the diffuser film produces an iridescent floating-text visual effect for displaying health metrics and device status without a conventional flat-panel display substrate."

3. Technical Justification and Novelty

Current wearable displays rely on rigid glass or plastic substrates (OLED/LCD) that create a distinct visual boundary (bezel) and limit the flexibility of the device. The proposed holographic micro-LED architecture offers several novel advantages:

- Seamless Integration:** By utilizing a diffuser film rather than a rigid screen, the display area becomes indistinguishable from the rest of the band when inactive.

2. **Enhanced Flexibility:** Micro-LED arrays can be mounted on flexible polyimide substrates, allowing the entire display region to bend naturally with the wrist.
3. **Iridescent Visuals:** The specific optical properties of the holographic diffuser film scatter the highly directional micro-LED light to create a unique “floating” effect, improving visibility in direct sunlight and offering a distinct aesthetic advantage.

4. Engineering Roadmap for CIP Implementation

While the CIP claim protects the concept immediately, the engineering realization is planned for V2/V3 hardware (2-3 year horizon).

1. **Phase 1 (Current):** Secure CIP filing date based on conceptual architecture and optical simulations.
2. **Phase 2 (Year 1):** Prototyping flexible micro-LED matrices (e.g., using technologies from Kura AR or Lumus adapted for macro-scale viewing).
3. **Phase 3 (Year 2):** Material science integration of the holographic diffuser film into the carbon fiber/silicone band manufacturing process to ensure durability and water resistance.
4. **Phase 4 (Year 3):** Commercialization in the “HEALTH-BAND Neuro Holographic Edition.”

5. Filing Strategy

The CIP must be filed before the one-year anniversary of the provisional application (May 23, 2027) to claim priority back to the original filing date for the shared subject matter (the Zero-Hole Architecture, sensors, etc.), while establishing a new priority date for the holographic display claims.